Course 301 Topic project – Using time series analysis for sales and demand forecasting

Skills	Criteria	Criterion has been met	
Importing packages and data	Load the appropriate libraries and models.	☐	
	Import both data files. Each file has four tabs, so ensure that you read the data from all four tabs.	☐	
Conducting initial data investigation	Note that the data provided is weekly data. If no sales happened in a particular week, there will be no data representation for that week. This means that the data is not at fixed intervals. As a result, resample the data and fill in missing values with 0, such that even weeks with 0 sales is represented.	☐	
	Convert the ISBNs to a string value.	☐	
	Convert date to datetime object. (Recall that setting the date as the index has several advantages for time series handling.)	☐	
	Filter out the ISBNs (from all four tabs) wherein sales data exists beyond 2024-07-01. Show all the ISBNs that satisfy this criterion. Capture this in your report.	☐	
	Plot the data of all the ISBNs from the previous step by placing them in a loop.	☐	
	Investigate these plots to understand the general sales patterns, and comment on the general patterns visible in the data. Do the patterns drastically change for the period 1–12 years vs the period 12–24 years? Explain why or why not with possible reasons.	☐	
	Select two books from the list (***The Alchemist*** and ***The Very Hungry Caterpillar***) for further analysis. Focus on the period >2012-01-01. Filter the sales data for both these books to retain the date range >2012-01-01, until the final datapoint.	☐	
Classical techniques	Perform decomposition on the data for both books. Determine what type of decomposition is suitable for each book, and comment on the components' characteristics.	☐	
	Perform ACF and PACF on both books. Comment on the results and what they indicate.	☐	
	Check for stationarity of the data for both books. Comment on the results and what they indicate.	☐	
	Perform Auto ARIMA on both books. **The forecast horizon is the final 32 weeks of data.** All prior data (from 2012-01-01 onwards) can be used as the training data. Set reasonable bounds for Auto ARIMA's various parameters so that Auto ARIMA can identify the most suitable model.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 32 weeks of the data
	Comment on the best model provided by Auto ARIMA for both books.	☐	
	Find the residuals of the 'best' model for both books. Comment on the residuals.	☐	
	Use the best model to predict the final 32 weeks of data. Plot the prediction along with the confidence intervals for both books.		
	Comment on how the prediction compares with the actual values.	☐	
Machine learning and deep learning techniques	Prepare the data to feed into the machine learning models. The forecast horizon is **32 weeks**. The training data consists of all prior data, up to 2012-01-01.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 32 weeks of the data
	Create the required pipeline for the XGBoost model.	☐	
	Perform cross-validation.	☐	
	Perform parameter tuning (including window_length) using grid search.	☐	
	Identify the best models.	☐	
	Use the best models to forecast the **final 32 weeks** of sales data for both books.	☐	
	Plot the original data along with the predictions.	☐	
	Display the MAE and MAPE.	☐	
	Create an LSTM model for both books.	☐	
	Apply KerasTuner, and perform hyperparameter tuning using the training data, for both books.	☐	

	Use the best model to predict the **final 32 weeks** of data for both books. Plot the prediction for both books.	☐	
	Plot the original data along with the predictions.	☐	
	Display the MAE and MAPE.	☐	
Hybrid model	Apply a hybrid model of SARIMA and LSTM in **sequential combination** wherein the residuals from SARIMA will be forecasted using LSTM. The final prediction will be the sum of the predictions from SARIMA and LSTM. The LSTM will be trained on the residuals obtained during the training of the SARIMA model. **The forecast horizon will be the final 32 weeks.** Use KerasTuner to get the best model. Plot the results. Display the MAE and MAPE, and comment on the results.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 32 weeks of the data
	Apply a hybrid model of SARIMA and LSTM in **parallel combination** wherein the predictions from SARIMA and the predictions from LSTM will be combined in the form of a weighted average. The final prediction will be the weighted sum of the predictions from SARIMA and LSTM. Both the SARIMA and the LSTM will be trained separately on the complete training set. **The forecast horizon will be the final 32 weeks.** Use KerasTuner to get the best model. Plot the results. Display the MAE and MAPE, and comment on the results.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 32 weeks of the data
	Modify the weightage in the parallel combination model to get different results. Find the weightage giving the best results, and comment on those results.	☐	
Monthly prediction	Instead of a weekly prediction, perform a monthly prediction. Aggregate the weekly sales data to monthly sales data for both books.	☐	
	Train the XGBoost model on this data. Forecast using the trained model. **The forecast horizon is 8 months.** Plot the results, and calculate the MAE and MAPE.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 8 months of the data
	Train the SARIMA model (using Auto ARIMA) on this data. Forecast using the trained model. **The forecast horizon is 8 months.** Plot the results, and calculate the MAE and MAPE.	☐	**For more clarity:** The training data includes all data from 2012-01-01, up to but not including the start of the forecasting horizon. The Forecast horizon is the final 8 months of the data
	Compare and contrast the monthly predictions of both books against the weekly predictions. Comment on which one is more accurate or if they are the same.	☐	
Report: Communicating business impact and insights	The report is between 800–1,000 words.	☐	
	The report documents the approach used.	☐	
	The report is clear, well-organised, and engaging to facilitate learning from the analysis.	☐	
	Conclusions drawn are clearly supported by the data.	☐	
	The code is well-organised and well-presented.	☐	
	The report captures and summarises the comments requested in earlier steps.	☐	
	The report consists of final insights, based on the output obtained from the various models employed.	☐	